KUNAL KUMAR

AI ENGINEER

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SKILLSUMMARY

Languages	Python, SQL
Frameworks	Matplotlib, Pandas, Numpy, Seaborn, FastAPI, Streamlit,TensorFlow, Keras, POSTMAN, MYSQL, REST API, SSMS, Tesseract-OCR, pygame(basics), pyautogui, astrapy, ngrok
TechnicalSkills	GenAI Solution using azure OpenAI, Computer Vision, Face-Recognition, Pose-Estimation, Virtual Try on Clothes, NLP, Text to speech, Speech To Text, Machine Learning, RAG, Prompt Engineering, Fine-Tuning LLMs, Langchain, Object Detection, Data Visualization, Tableau, Power-BI(basics), Predictive Analysis, Random Forest, Decision Tree, Transfer Learning, CNN, GANs, Transformers, VAEs, Fine-TuningLlama2-7b

WORKEXPERIENCE

AIENGINEER

APPseCONNECT

- **Automated the Minutes of Meeting Generation Process and Proposal Generation Process** for the Sales and Marketing Team by **fetching Teams Transcripts using** Microsoft Graph API using **SDK.** With this **Removed** the **Manual Task** needed.

- **Created Process-Flow to automate fetching the Top Trending Topics on the Internet** And then **Generating Marketing Contents** based on those**.**

- **Created UI using Streamlit** for taking prompt, storing **word-Embeddings on Astradb vector database** and doing **Similarity Search** to fetch the most similar article and then generating accurate STEPS TO SOLVE.

- **Copilot App** to be installed on APPseCONNECT portal **for Support Engineers/End-Users.**
- **Created API** for Web-Application using **FastAPI and Uvicorn** and then hosted the api server on internet using **ngrok** and created 2FA system with password to make it secure.

PROJECTs

1. **COPILOT App** to explain the **ERROR EXPLANATION and STEPS TOSOLVE** to the APPseCONNECT End-Users **in real time** during process-flow data exchange happening between any 2 Apps like ERP and CRM.
   ● Took Each Node description, Node Names, and Activity Log for each node involved in the Process-Flow from **Snapshot.**
   ● Pre-Processed data to **decrease the Total Tokens by removing the Junk-Data.**

● **Created a UI using Streamlit** to take user-input prompt in the form of org_id and snapshot_id and then made GPT-4 to give response having PROCESS-FLOWGOAL,ERRORDETAILS,POSSIBLEREASONS,STEPSTOSOLVE
In a **Non-Technical language.**

2. Restricted Area Surveillance System | LINK

- Used **YOLOv8** model for **object detection**.
- Using face-recognition pre-trained model performed **face-detection, face-recognition**
- Improvements done:
    1. **Added audio** to the output which is not present in output of yolov8.
    2. **Created Alarm system** and **Email sending system using smtplib**.

3. **Built GANS(Generator and Discriminator)** from scratch using CNN with 200 epochs to generate realistic but fake human pictures artificially. Result were poor due to **lower number of epochs** and **lower picture resolution** used to train the model(due to **hardware constraints**). In future, will try to improve the output picture quality.

4. Speech to Speech Chatbot for Airplane flying school | LINK

- Performed Text Processing to create Question Answer pair Template
- Used these Tools and technologies **Transformers, Llama-2-7b, NLP,** PyTorch, **Hugging-Face , PEFT, SFT**
- Fine-Tuned Llama-2-7b LLM using hugging face API on Airplane Flying Handbook dataset of US Federal Aviation Administration(DOT-FAA).

- Took prompts in speech medium then converted **speech to text** and then passed the text to our customized chatbot. The chatbot gave response in text form which I converted it into speech medium using **Text to Speech** technology.

5. Pneumonia Detection | LINK

- Built a **CNN** network to classify **chest X-ray** images as pneumonia-affected or normal.
- Achieved**75.66%accuracy**.Plan to improve accuracy by optimizing image size and model parameters.
- Used theseTools andTechnology: CNN, Deep earning, Transfer Learning, Keras, Tensorflow, **VGG-16**

6. Automation solution for Job Application Filling Process |LINK

- Extracted text from Resume PDF using **PyMuPDF**.
- Used **Regular Expressions to** extract Email, Name, Phone Number, Skills from the Resume
- Extracted details were uploaded to the **UI created by me** to fill applicant details using requests.

- Performed this same task using Neural Network based Transformers using Pre-Trained **BERTLLM** for Named Entity Recognition. However result was not good.

- Outcome: Regular Expressions gave far better output than transformers.

7. Restaurant Revenue Prediction

- This project was part of the hackathon organized by Great Learning platform.
  a. Performed Data Cleaning and Exploratory Data Analysis and then built a **RandomForest Regressor** ML model
  b. I secured **05 Rank** in this Hackathon,with **90.5% accuracy**.
  c. However, after the competition was over, I **increased the model's accuracy t o 95.5%** using **SMOTE(for Balancing the data)** and XG-Boost model.

8. Smart App for Online Teaching Industry |LINK

- Used **Mediapipe** to do **Background Removal**.
- Used **pyautogui** for **taking screenshots** of each frame.
- Used **OpenCV** to combined both and display it in Real-Time so that the teacher will be seen speaking on bottom right corner of his laptop screen due to background removal and laptop screen addition.

9. Children's Music Device | LINK

- Used **Pygame** to create UI to take input from user and Then execute the code based on that.
- Gave **3 songs to choose**.Only one song would be played at a time after the child chooses any one song.

10. **QR CODE and BARCODE Generation** using Python. Generated slogans for a New Restaurant using **GenAI** and then Created QR CODE having this slogan as stored information

11. **SNAPCHAT like App** for users to have fun and entertainment with funny facial expressions, ghost presence, body skeleton of users on live video feed, background removal and addition, etc

Some Extra Projects:

1. **2 Factor Authentication** System using SMTPLIB|**LINK**
2. **Pose Estimation** using **Mediapipe** | **LINK**
3. **Virtual Try of clothes**. Currently Result is very poor, have to improve it | **LINK**
4. Used **Diffusion Models like Stable Diffusion** and OpenAI's **DALLE**.
5. Used **WHISPER to convert office meeting voice recordings into text form using POSTMAN.**

EDUCATION

- a. **Post Graduation in Artificial Intelligence and Machine Learning** (**Undergoing**)University of Texas at Austin

- b. **B-Tech in Electronics and Communication** Engineering 2014-2018(Vellore Institute of Technology (VIT Vellore))